Time	User	Utterance
03:44	Old	I dont run graphical ubuntu, I run ubuntu server.
03:45	kuja	Taru: Haha sucker.
03:45	Taru	Kuja: ?
03:45	bur[n]er	Old: you can use "ps ax" and "kill (PID#)"
03:45	kuja	Taru: Anyways, you made the changes right?
03:45	Taru	Kuja: Yes.
03:45	LiveCD	or killall speedlink
03:45	kuja	Taru: Then from the terminal type: sudo apt-get update
03:46	_pm	if i install the beta version, how can i update it when the final version comes out?
03:46	Taru	Kuja: I did.

Sender	Recipient	Utterance
Old		I dont run graphical ubuntu, I run ubuntu server.
bur[n]er	Old	you can use "ps ax" and "kill (PID#)"
kuja	Taru	Haha sucker.
Taru	Kuja	?
kuja	Taru	Anyways, you made the changes right?
Taru	Kuja	Yes.
kuja	Taru	Then from the terminal type: sudo apt-get update
Taru	Kuja	I did.

Conversation Modeling

- Shang et al. (2015) [44]
- Vinyals et al. (2015) [45]
- Lowe et al. (2015) [46]
- Dodge et al. (2015) [47]
- Weston et al. (2016) [48]
- Serban et al. (2016) [49]
- Bordes et al. (2017) [50]
- Serban et al. (2017) [51]

Task 1: Single Supporting Fact
Mary went to the bathroom.
John moved to the hallway.
Mary travelled to the office.
Where is Mary? A:office

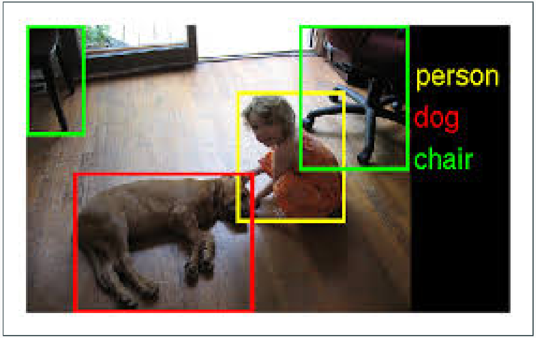
Task 2: Two Supporting Facts
John is in the playground.
John picked up the football.
Bob went to the kitchen.
Where is the football? A:playground

Task 3: Three Supporting Facts
John picked up the apple.
John went to the office.
John went to the kitchen.
John dropped the apple.
Where was the apple before the kitchen? A:office

Task 4: Two Argument Relations
The office is north of the bedroom.
The bedroom is north of the bathroom.
The kitchen is west of the garden.
What is north of the bedroom? A: office
What is the bedroom north of? A: bathroom

Question Answering

- Hermann et al. (2015) [52]
- Chen et al. (2016) [53]
- Xiong et al. (2016) [54]
- Seo et al. (2016) [55]
- Dhingra et al. (2017) [56]
- Wang et al. (2017) [57]
- Hu et al. (2017) [58]



Object Detection/Recognition

- Semantic Segmentation (Long et al, 2015) [59]
- Recurrent CNNs (Liang et al., 2015) [60]
- Faster RCNN (Ren et al., 2015) [61]
- Inside-Outside Net (Bell et al., 2015) [62]
- YOLO9000 (Redmon et al., 2016) [63]
- R-FCN (Dai et al., 2016) [64]
- Mask R-CNN (He at al., 2017) [65]
- Video Object segmentation (Caelles et al., 2017) [66]

Visual Tracking

- Choi et al. (2017) [67]
- Yun et al. (2017) [68]
- Alahi et al. (2017) [69]

1. Top view of the lights of a city at night, with a well-illuminated square

1. Tourists are sitting at a long table with beer bottles on it in a rather dark restaurant

[redacted]

Image Captioning

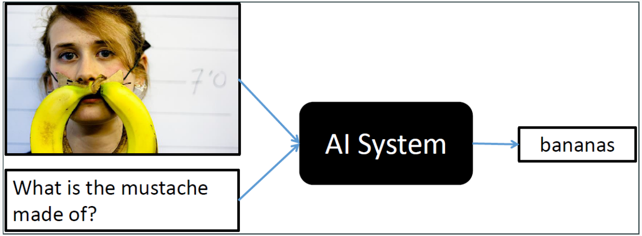
- Mao et al. (2014) [70]
- Mao at al. (2015) [71]
- Kiros et al. (2015) [72]
- Donahue et al. (2015) [73]
- Vinyals et al. (2015) [74]
- Karpathy et al. (2015) [75]
- Fang et al. (2015) [76]
- Chen et al. (2015) [77]

A group of young men playing a game of soccer

A man riding a wave on top of a surfboard.

Video Captioning

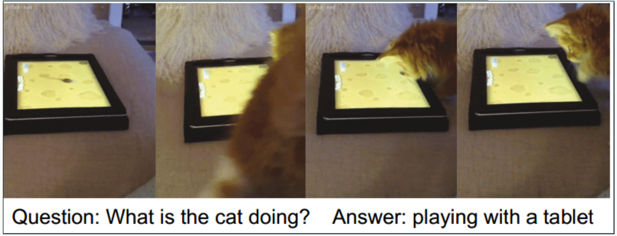
- Donahue et al. (2014) [78]
- Venugopalan at al. (2014) [79]
- Pan et al. (2015) [80]
- Yao et al. (2015) [81]
- Rohrbach et al. (2015) [82]
- Zhu et al. (2015) [83]
- Cho et al. (2015) [34]



Visual Question Answering

- Santoro et al. (2017) [84]
- Hu at al. (2017) [85]
- Johnson et al. (2017) [86]
- Ben-younes et al. (2017) [87]
- Malinowski et al. (2017) [88]
- Kazemi et al. (2016) [89]

Video Question Answering



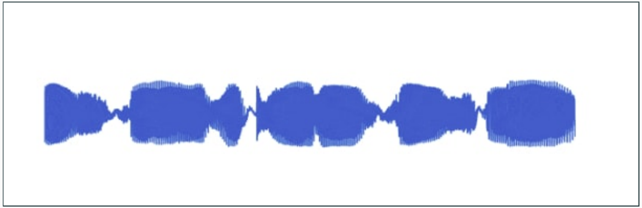
- Tapaswi et. al. 2016 [90]
- Zeng et. al. 2016 [91]
- Maharaj et. al. 2017 [92]
- Zhao et. al. 2017 [93]
- Yu Youngjae et. al. 2017 [94]
- Xue Hongyang et. al. 2017 [95]
- Mazaheri et. al. 2017 [96]

Video Summarization

- Chheng 2007 [97]
- Ajmal 2012 [98]
- Zhang Ke 2016 [99]
- Zhong Ji 2017 [100]
- Panda 2017 [101]

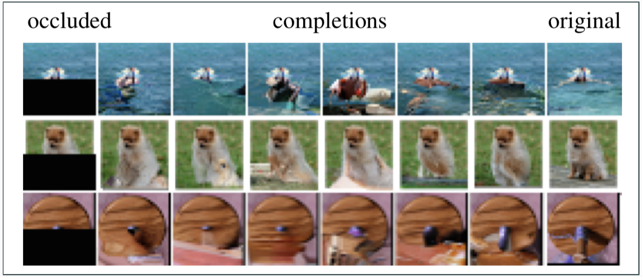
Generating Authentic Photos

- Variational Autoencoders (Kingma et. al., 2013) [102]
- Generative Adversarial Networks (Goodfellow et. al., 2014) [103]
- Plug & Play generative nets (Nguyen et al., 2016) [104]
- Progressive Growing of GANs (Karras et al., 2017) [105]



Generating Raw Audio

- Wavenets (Oord et. al., 2016) [106]



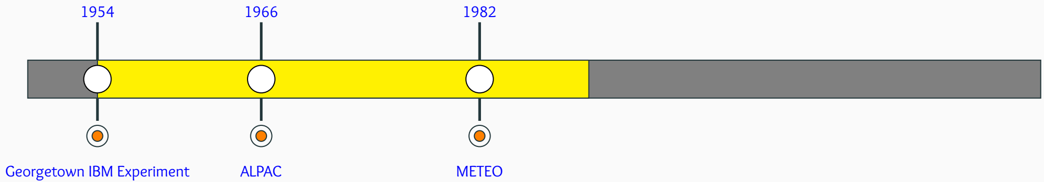
Pixel RNNs

- (Oord et al., 2016) [107]
- (Oord et al., 2016) [108]
- (Salimans et al., 2017) [109]

Chapter 9: The Rise of the Transformers

Rule Based Systems

Intial Machine Translation Systems used hand crafted rules and dictionaries to translate sentences between few politically important language pairs (e.g., English -Russian). They could not live upto the initial hype and were panned by the ALPAC report (1966)



Statistical MT

The IBM Models for Machine Translation gave a boost to the idea of data driven statistical NLP which then ruled NLP for the next 2 decades till Deep Learning took over!

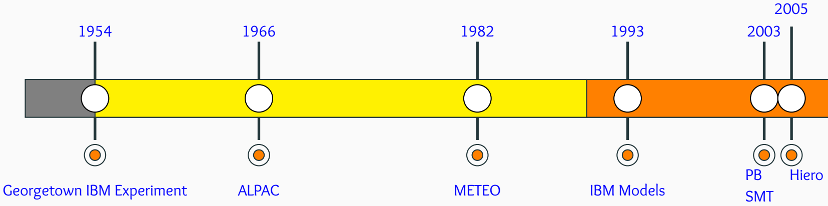
The Mathematics of Statistical Machine Translation: Parameter Estimation

Peter F. Brown*
IBM T.J. Watson Research Center

Stephen A. Della Pietra*
IBM T.J. Watson Research Center

Vincent J. Della Pietra*
IBM T.J. Watson Research Center

Robert L. Mercer*
IBM T.J. Watson Research Center



Neural MT

The introduction of seq2seq models and attention [35] (perhaps, the idea of the decade!) lead to a paradigm shift in NLP ushering the era of bigger, hungrier (more data), better models!

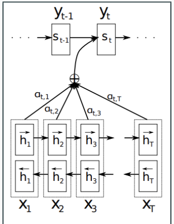
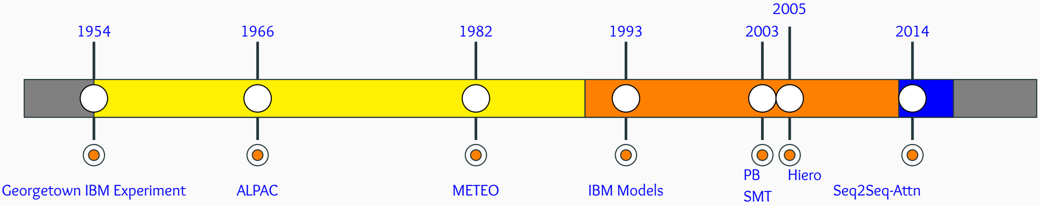


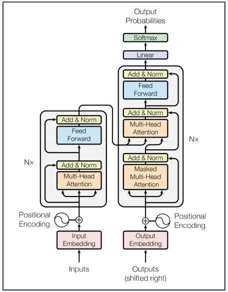
Figure 1: The graphical illustration of the proposed model trying to generate the t-th target word y_t given a source sentence $(x_1, x_2, \ldots, x_T)$.

Source: Bahdanau et. al. [35]

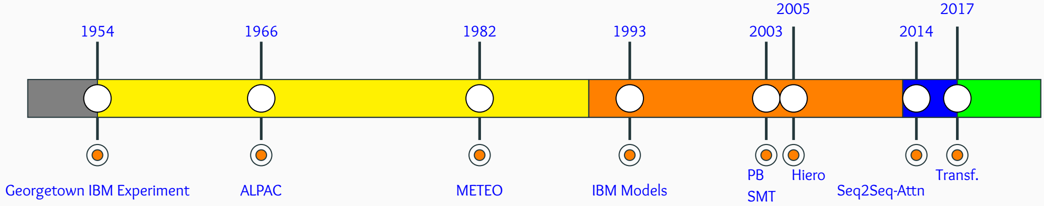


The Transformer Revolution

It is rare for a field to see two dramatic paradigm shifts in a short span of 4 years! Since their inception transformers have taken the NLP world by storm leading to the development of insanely big models trained on obscene amounts of data!

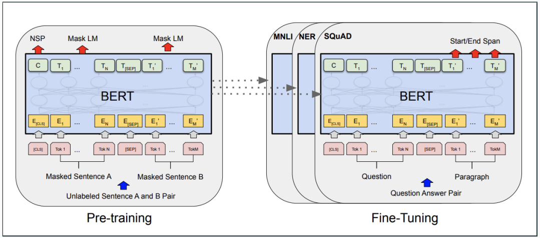


Source: Vaswani et. al. [110]

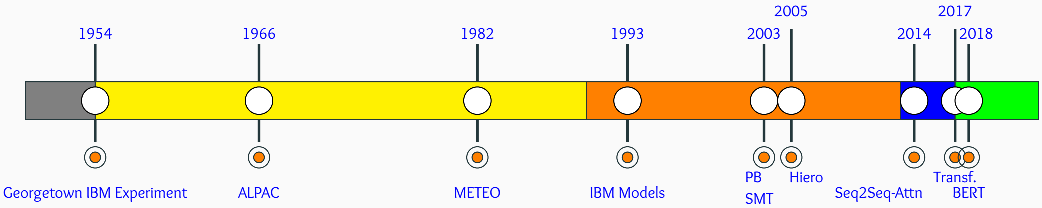


The Transformer Revolution

Most NLP applications today are driven by BERT and its variants. The key idea here was to learn general langauge characteristics using large amounts of unlabeled corpora and then fine-tune the model for specific downstream tasks.



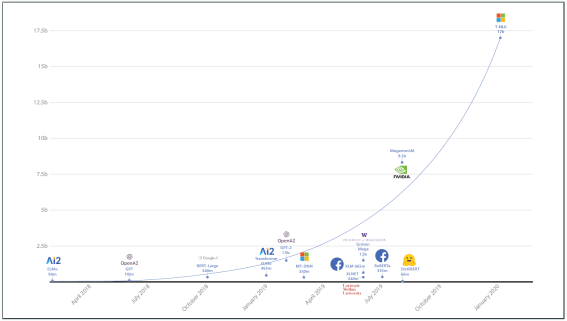
Source: Devlin et. al.[111]

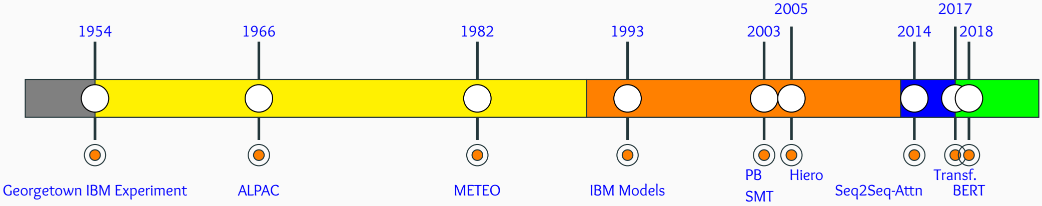


The Billion Parameter Club

The models are becoming bigger and bigger and bigger!

Source: https://msturing.org/





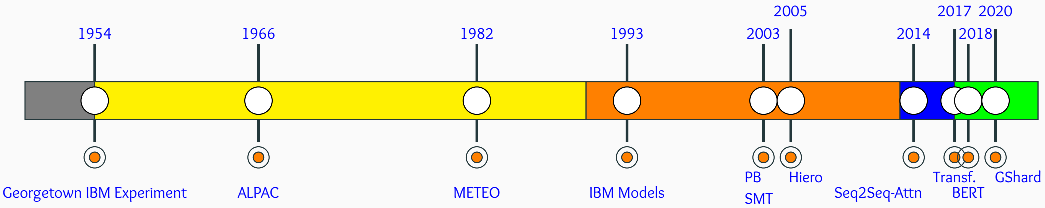
The Trillion Parameter Club

Trained on 100 languages, with a total of 13B examples, 1 Trillion Parameters on 2048 TPUs!

This is insane!

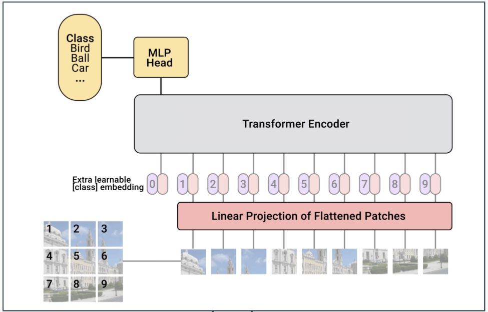
GShard: Scaling Giant Models with Conditional Computation and Automatic Sharding

Dmitry Lepikhin lepikhin@google.com
HyoukJoong Lee hyouklee@google.com
Yuanzhong Xu yuanzx@google.com
Dehao Chen dehao@google.com
Orhan Firat orhanf@google.com
Yanping Huang huangyp@google.com
Maxim Krikun krikun@google.com
Noam Shazeer noam@google.com
Zhifeng Chen zhifengc@google.com

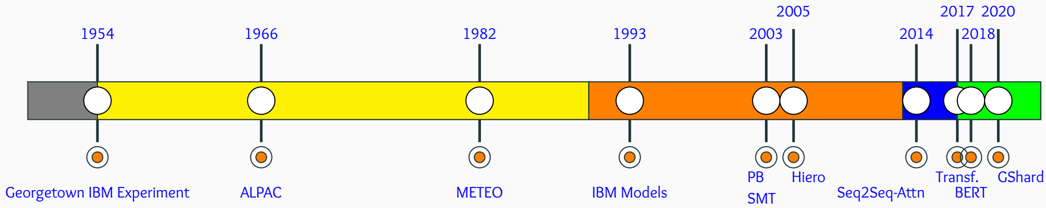


From Language To Vision

A vision model[a] based as closely as possible on the Transformer architecture originally designed for text-based tasks (another paradigm shift from CNNs which have been around since 1980s!)



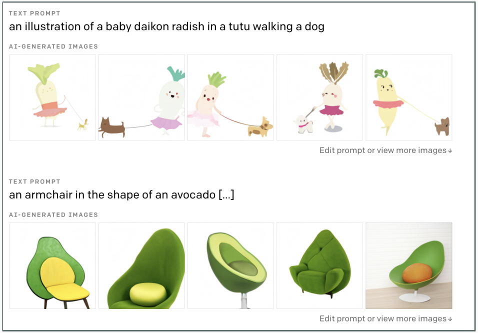
[a] Source:https://ai.googleblog.com/2020/12/transformers-for-image-recognition-at.html

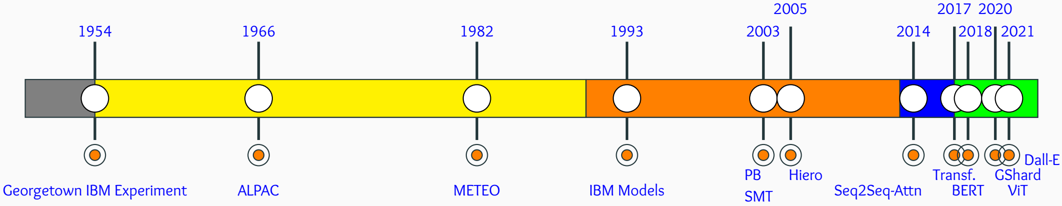


From Language To Vision

DALL·E[a] is a 12-billion parameter version of GPT-3 trained to generate images from text descriptions, using a dataset of text–image pairs.

[a] https://openai.com/blog/dall-e/

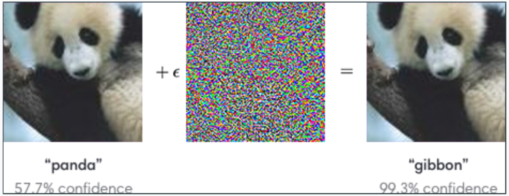




Chapter 10: Calls for Sanity (Interpretable, Fair, Responsible, Green AI)

The Paradox of Deep Learning

Why does deep learning work so well despite

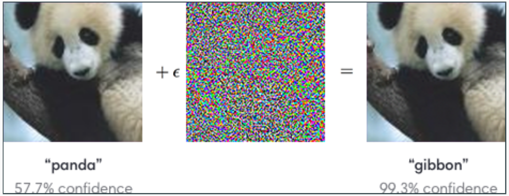


*https://arxiv.org/pdf/1710.05468.pdf

The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)



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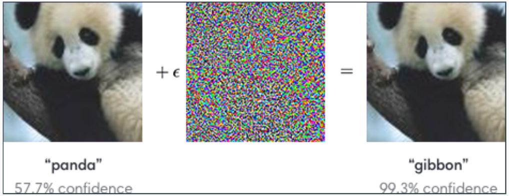
*https://arxiv.org/pdf/1710.05468.pdf

The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)
- numerical instability (vanishing/exploding gradients)



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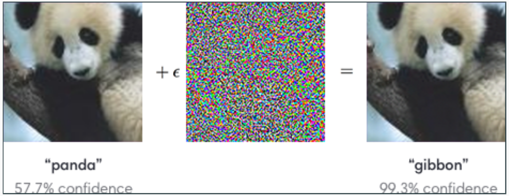
The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)
- numerical instability (vanishing/exploding gradients)
- sharp minima (leading to overfitting)



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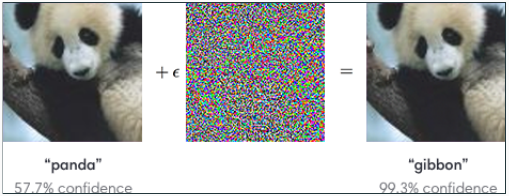
The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)
- numerical instability (vanishing/exploding gradients)
- sharp minima (leading to overfitting)
- non-robustness (see figure)



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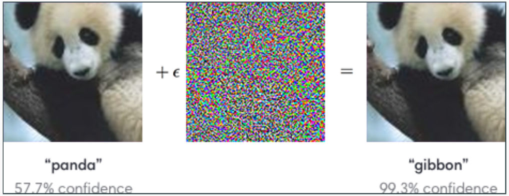

No clear answers yet but ...

---

*https://arxiv.org/pdf/1710.05468.pdf

The Paradox of Deep Learning

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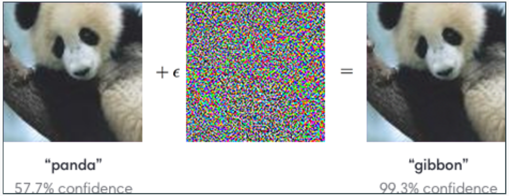

No clear answers yet but ...

- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!*

---

*`https://arxiv.org/pdf/1710.05468.pdf`

The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)
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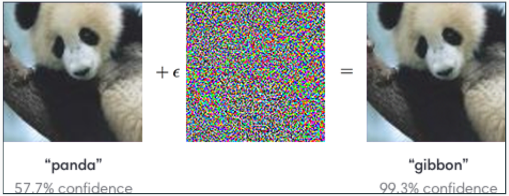


No clear answers yet but ...

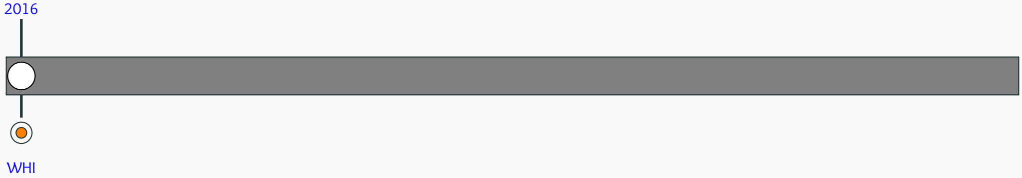
- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!*
- Hopefully this will bring sanity to the proceedings !

*https://arxiv.org/pdf/1710.05468.pdf

Tell me why!

Workshop on Human Interpretability in Machine Learning

We still do not know much about why DL models do what they do!



Tell me why!

Clever Hans was a horse that was supposed to be able to do lots of difficult mathematical sums and solve complicated problems. Turns out, it was giving the right answers by watching the reactions of the people watching him.

A repository to benchmark machine learning systems' vulnerability to adversarial examples.

2016

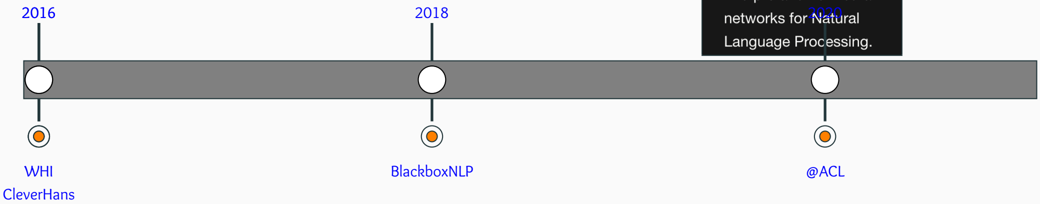
WHI
CleverHans

Tell me why!

Push for analyzing and interpreting neural networks for NLP

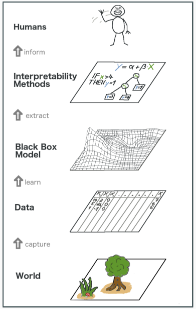
Analyzing and interpreting neural networks for NLP

Revealing the content of the neural black box: workshop on the analysis and interpretation of neural networks for Natural Language Processing.

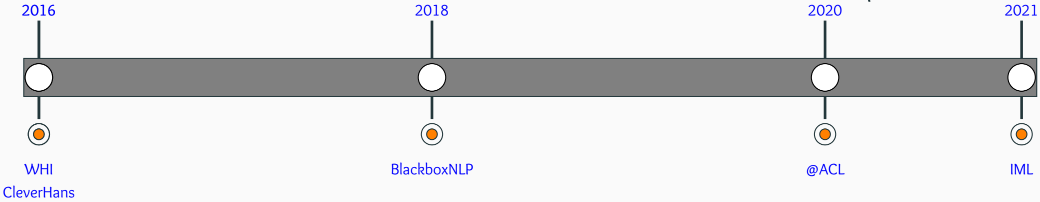


Tell me why!

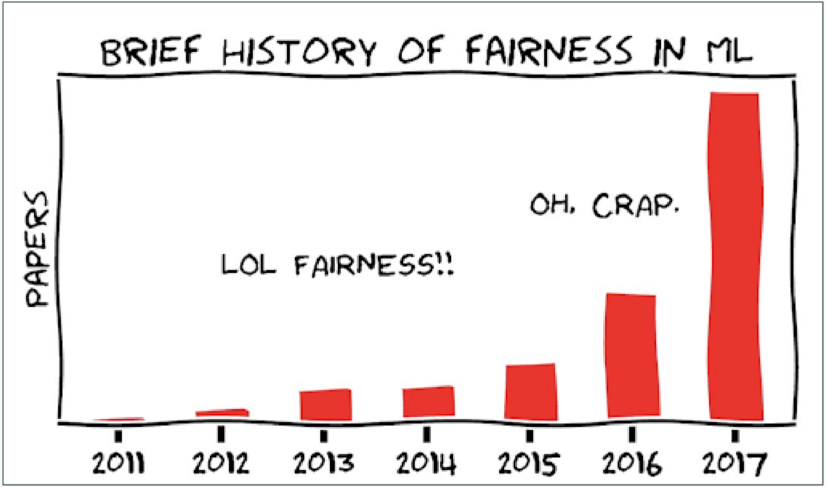
Interpretable Machine Learning: A Guide for Making Black Box Models Explainable. – Christoph Molnar



Source: IML: Christoph Molnar



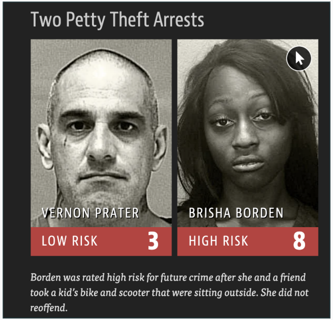
Be Fair and Responsible!



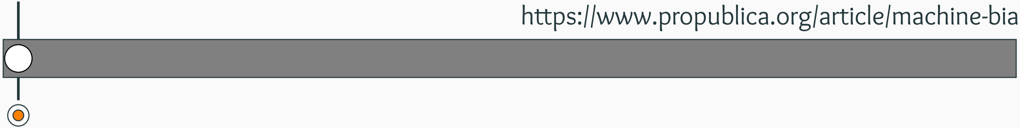
Source: https://fairmlclass.github.io/ (Moritz Hardt)

Be Fair and Responsible!

“There’s software used across the country to predict future criminals. And it’s biased against blacks.” - Propublica

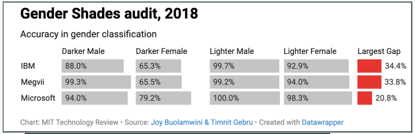


Source:
https://www.propublica.org/article/machine-bias-



Be Fair and Responsible!

"Facial Recognition Is Accurate, if You're a White Guy" - MIT Media



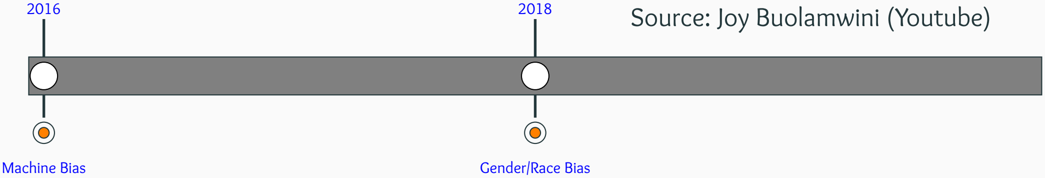
Gender Shades audit, 2018

Accuracy in gender classification

	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
IBM	88.0%	65.3%	99.7%	92.9%	34.4%
Megvii	99.3%	65.5%	99.2%	94.0%	33.8%
Microsoft	94.0%	79.2%	100.0%	98.3%	20.8%

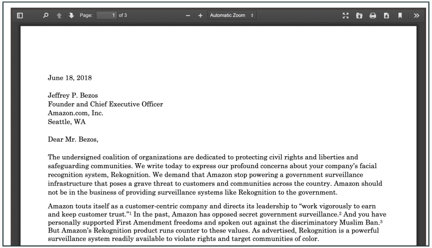
Chart: MIT Technology Review • Source: Joy Buolamwini & Timnit Gebru • Created with Datawrapper

Source: Joy Buolamwini (Youtube)



Be Fair and Responsible!

In 2018, nearly 70 civil rights and research organizations wrote a letter to Jeff Bezos demanding that Amazon stop providing face recognition technology to governments.



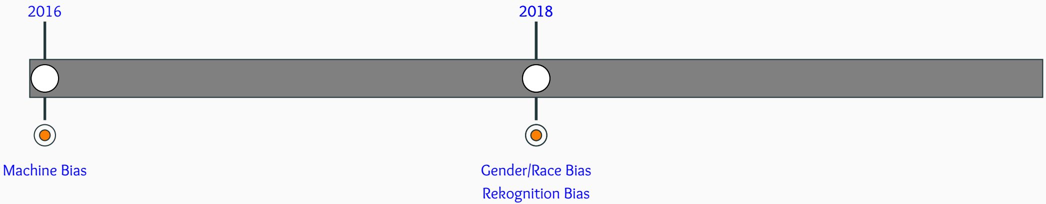
June 18, 2018

Jeffrey P. Bezos
Founder and Chief Executive Officer
Amazon.com, Inc.
Seattle, WA

Dear Mr. Bezos,

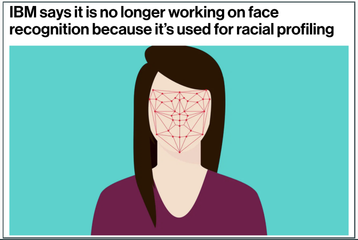
The undersigned coalition of organizations are dedicated to protecting civil rights and liberties and safeguarding communities. We write today to express our profound concerns about your company's facial recognition system, Rekognition. We demand that Amazon stop powering a government surveillance infrastructure that poses a grave threat to customers and communities across the country. Amazon should not be in the business of providing surveillance systems like Rekognition to the government.

Amazon touts itself as a customer-centric company and directs its leadership to "work vigorously to earn and keep customer trust."[1] In the past, Amazon has opposed secret government surveillance.[2] And you have personally supported First Amendment freedoms and spoken out against the discriminatory Muslim Ban.[3] But Amazon's Rekognition product runs counter to these values. As advertised, Rekognition is a powerful surveillance system readily available to violate rights and target communities of color.



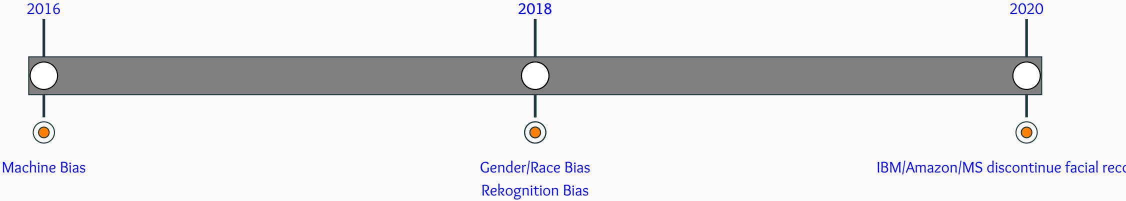
Be Fair and Responsible!

Microsoft refuses to sell police its facial-recognition technology, following similar moves by Amazon and IBM


IBM says it is no longer working on face recognition because it's used for racial profiling

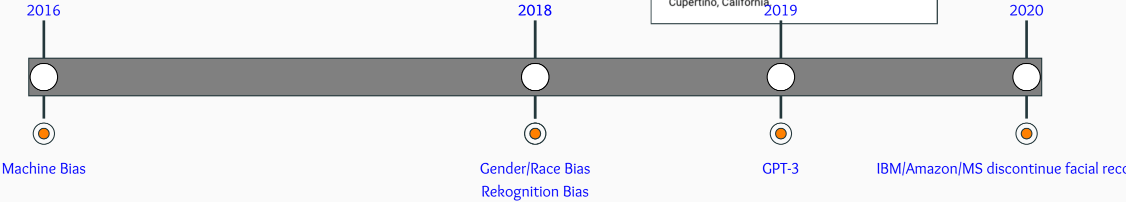

Technology

Microsoft won't sell police its facial-recognition technology, following similar moves by Amazon and IBM



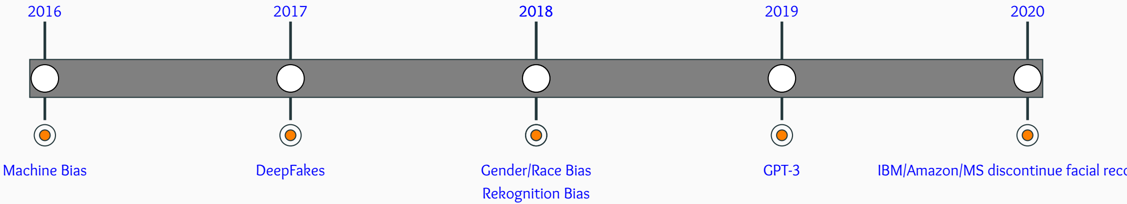
Be Fair and Responsible!

“Due to our concerns about malicious applications of the technology, we are not releasing the trained model.” – – *OpenAI*



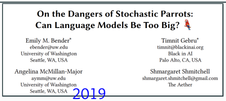
Be Fair and Responsible!

What started off as an innocuous project for mimicking facial expressions has since lead to many apps and creation of fake videos for blackmailing, pronography and swaying elections!



Be Fair and Responsible!

"Models are only as good as the data. Be responsible while curating data." – *Bender et. al.*



On the Dangers of Stochastic Parrots:
Can Language Models Be Too Big? 🦜

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Timnit Gebru*, timnit@blackinai.org, Black in AI, Palo Alto, CA, USA

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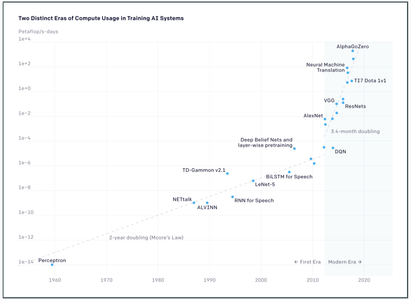
- 2016 – Machine Bias
- 2017 – DeepFakes
- 2018 – Gender/Race Bias, Rekognition Bias
- 2019 – GPT-3
- 2020 – IBM/Amazon/MS discontinue facial reco…, Stochastic Parrots!

Push for Green AI

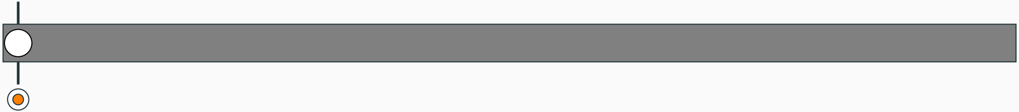
The computations required for deep learning research have been doubling every few months, resulting in an estimated 300,000x increase from 2012 to 2018 – *AllenAI*

Ironically, deep learning was inspired by the human brain, which is remarkably energy efficient.



https://openai.com/blog/ai-and-compute/

2019



GreenAI

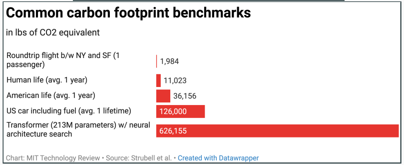
Push for Green AI

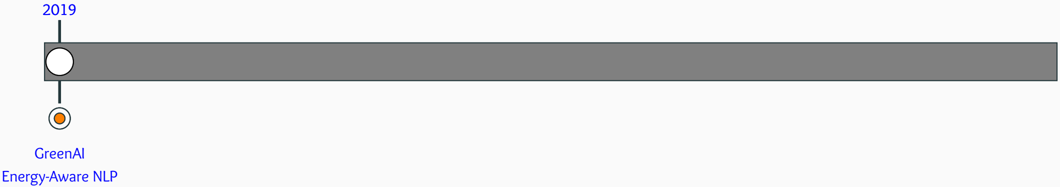
Call for energy and policy considerations for Deep Learning

	Date of original paper	Energy consumption (kWh)	Carbon footprint (lbs of CO2e)	Cloud compute cost (USD)
Transformer (65M parameters)	Jun, 2017	27	26	$41-$140
Transformer (213M parameters)	Jun, 2017	201	192	$289-$981
ELMo	Feb, 2018	275	262	$433-$1,472
BERT (110M parameters)	Oct, 2018	1,507	1,438	$3,751-$12,571
Transformer (213M parameters) w/ neural architecture search	Jan, 2019	656,347	626,155	$942,973-$3,201,722
GPT-2	Feb, 2019	-	-	$12,902-$43,008

Note: Because of a lack of power draw data on GPT-2's training hardware, the researchers weren't able to calculate its carbon footprint.

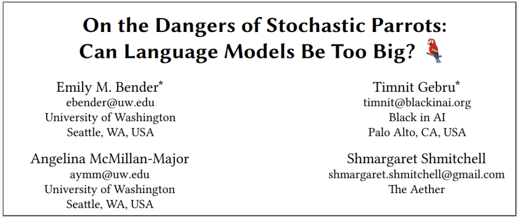
Table: MIT Technology Review • Source: Strubell et al. • Created with Datawrapper





Push for Green AI

"Is it fair that the residents of the Maldives (likely to be underwater by 2100) or the 800,000 people in Sudan affected by drastic floods pay the environmental price of training and deploying ever larger English LMs, when similar large-scale models aren't being produced for Dhivehi or Sudanese Arabic?" – *Bender et. al.*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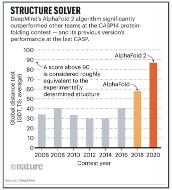
The Aether

Chapter 11: The AI revolution in Scientific Research (exciting times ahead!)

Accelerating Scientific Discovery[a]



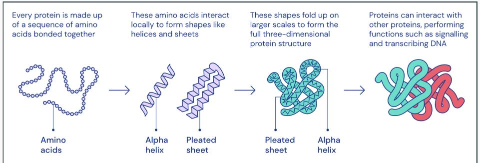
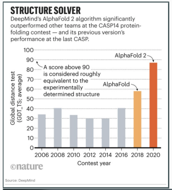
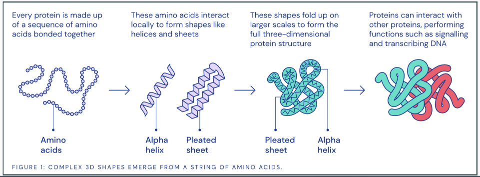


FIGURE 1: COMPLEX 3D SHAPES EMERGE FROM A STRING OF AMINO ACIDS.

[a] https://deepmind.com/blog/article/AlphaFold-Using-AI-for-scientific-discovery
https://ocean.org/stories/spotting-seals-from-space
https://www.quantamagazine.org/how-artificial-intelligence-is-changing-science-20190311/

Accelerating Scientific Discovery[a]





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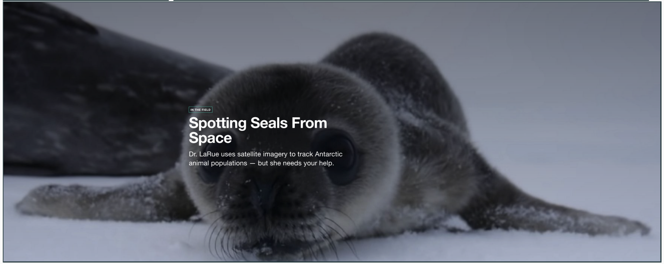

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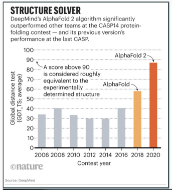
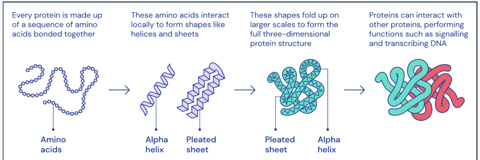
Accelerating Scientific Discovery[a]





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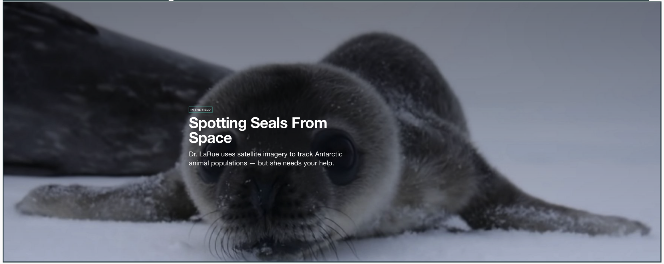
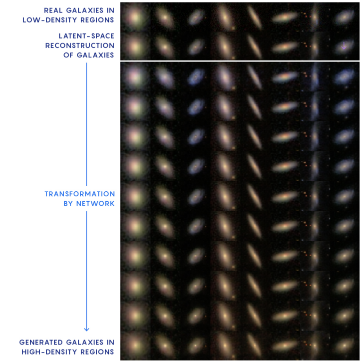


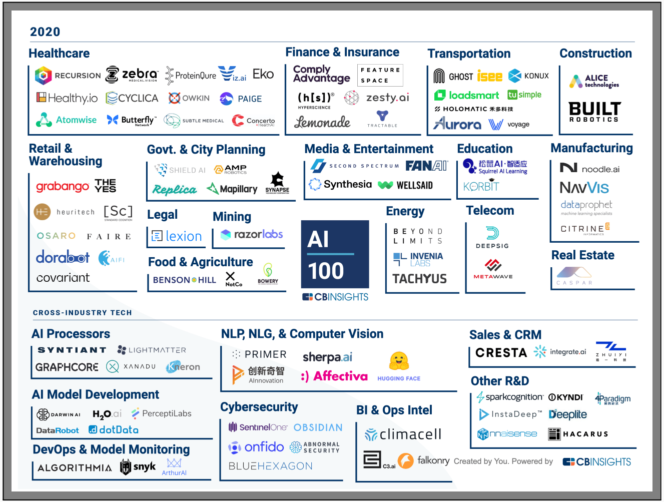


Using generative modeling, astrophysicists could investigate how galaxies change when they go from low-density regions of the cosmos to high-density regions, and what physical processes are responsible for these changes.

Adapted from K. Schawinski et al.; Source doi: 10.1051/0004-6361/201833800

[a] https://deepmind.com/blog/article/AlphaFold-Using-AI-for-scientific-discovery
https://ocean.org/stories/spotting-seals-from-space
https://www.quantamagazine.org/how-artificial-intelligence-is-changing-science-20190311/

https://github.com/ChristosChristofidis/awesome-deep-learning



Source: https://www.cbinsights.com/research/artificial-intelligence-top-startups/

[i] Source: https://www.cbinsights.com/research/artificial-intelligence-top-startups/

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